Verification (added). (i) The per-row $\log(\vee/k^+)$ extractions of this file were checked against the verified Group results where the comparison was run (e.g. Group III Row I) and agree exactly. (ii) The final adjoint-index BFKL expression was checked against its colour-singlet form at the end of the file: substituting $\langle\rho^b(x')\rho^c(x)\rangle \to \delta^{bc}\,\mu^2(x',x)/(N_c^2-1)$, the two agree *exactly* (relative difference $\sim 10^{-15}$, checked with exact SU(2) and SU(3) adjoint Wilson lines) – provided the traces in the singlet form are read as *adjoint* traces (adjoint U, $(T^e)_{ab} = -if^{eab}$). No corrections were needed in this file.

BFKL evolution of the projectile

1 From $4\,\rho$'s

1.1 From four ρ's

$$\frac{d^3N_{a\,\mathrm{NLO}}}{d^3k} = -\frac{i}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\log\left[\frac{\vee}{k^+}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[$$
$$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^a(y)\rho^c(x)\,U^{ed}(w')U^{eb}(w)\,\alpha$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^e(x')\rho^a(y)\rho^c(x)\,U^{db}(w)U^{de}(x')\,\alpha$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^c(x')\rho^a(y)\rho^e(x)\,U^{db}(w')U^{de}(x)\,\alpha'$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^c(x')\rho^a(y)\rho^e(x)U^{db}(w')U^{de}(w)\,\alpha'\Bigg] \tag{1.1}$$

$$\frac{d^3N_{b\,\mathrm{NLO}}}{d^3k} = \frac{i}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\log\left[\frac{\vee}{k^+}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^h(y)\rho^e(x)U^{bd}(w'U^{ce}(xU^{ah}(y)\,\alpha$$
$$+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)U^{cd}(x'U^{be}(wU^{ah}(y)\,\alpha'$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)U^{cd}(x'U^{be}(x)U^{ah}(y)\,\alpha'$$
$$+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^e(x')\rho^h(y)\rho^d(x)U^{cd}(xU^{be}(x')U^{ah}(y)\,\alpha\Bigg] \tag{1.2}$$

1.2 From $4\,\rho's$

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
&- \frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ba}(w')\,U^{bc}(w)\,\rho^a(x')\,\rho^c(x) \\
&+ \frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&- \frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ba}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x) \\
&+ \frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+ \frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&- \frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ba}(w')\,U^{bc}(w)\,\rho^a(x')\,\rho^c(x) \\
&- \frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ba}(w)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x') \\
&+ \frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hd}(w')\,U^{hb}(w)\,\rho^e(x')\,\rho^c(x) \\
&+ \frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x) \\
&- \frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x) \\
&- \frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{be}(w')\,U^{cl}(x)\,U^{ad}(x')\,\rho^h(x')\,\rho^l(x) \\
&+ \frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{cl}(x')\,U^{be}(w)\,U^{ad}(x)\,\rho^l(x')\,\rho^h(x) \\
&- \frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{hb}(w)\,\rho^a(x)\,\rho^e(x') \\
&+ \frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{hd}(x')\,\rho^a(x)\,\rho^e(x') \\
&- \frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hb}(w')\,U^{hd}(x')\,\rho^a(x)\,\rho^e(x') \\
&+ \frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hb}(w')\,U^{hd}(w)\,\rho^a(x)\,\rho^e(x') \\
&- \frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{be}(w')\,U^{cd}(x')\,U^{al}(x)\,\rho^l(x)\,\rho^h(x') \\
&+ \frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ce}(x')\,U^{bd}(w)\,U^{al}(x)\,\rho^h(x')\,\rho^l(x) \\
&- N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Bigg]
\end{aligned}
\tag{1.3}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Bigg[\\
+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
-3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ha}(z)\,U^{lb}(w)\,U^{hd}(x)\,U^{lm}(w')\,\rho^m(x')\,\rho^e(x) \\
+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ha}(z)\,U^{lb}(x)\,U^{hd}(x)\,U^{lm}(w')\,\rho^m(x')\,\rho^e(x) \\
-3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ha}(w)\,U^{lb}(x)\,U^{ld}(z)\,U^{hm}(w')\,\rho^m(x')\,\rho^e(x) \\
+3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ha}(z)\,U^{ld}(w)\,U^{lm}(w')\,U^{hb}(x)\,\rho^m(x')\,\rho^e(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(w)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x) \\
+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ha}(z)\,U^{lb}(x)\,U^{hd}(x)\,U^{lm}(x')\,\rho^m(x')\,\rho^e(x) \\
-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(w')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x) \\
-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ha}(w)\,U^{hl}(x')\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{hl}(x')\,U^{ma}(x)\,U^{md}(z)\,\rho^l(x')\,\rho^e(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{hb}(x)\,U^{el}(x')\,\rho^l(x')\,\rho^c(x) \\
+3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{ha}(x)\,U^{dl}(w')\,U^{eb}(z)\,\rho^l(x')\,\rho^c(x) \\
+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{md}(x')\,\rho^l(x)\,\rho^e(x') \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{md}(x')\,\rho^l(x)\,\rho^e(x') \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{mb}(x')\,\rho^l(x)\,\rho^e(x') \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{mb}(x')\,\rho^l(x)\,\rho^e(x') \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x) \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x) \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x) \\
+\frac{3}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ba}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x) \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(w)\,U^{hl}(x')\,U^{da}(x)\,\rho^l(x')\,\rho^c(x) \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x) \\
+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(x)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{hd}(x)\,\rho^e(x)\,\rho^c(x') \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(x')\,U^{hl}(x)\,U^{da}(z)\,\rho^l(x)\,\rho^c(x') \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(z)\,U^{cl}(x')\,U^{be}(x)\,\rho^h(x)\,\rho^l(x')\Bigg]
\end{aligned}
\tag{1.4}
$$

2 Group I (with c.c.)

2.1 Row I

NA

2.2 Row II

$$\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{i\,g^4\,f^{hbc}}{32\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x',z}\frac{(x'-w')^i}{(x'-w')^2}\left[\frac{(x-w)^i}{(x-w)^2}\,\frac{(w-z)\cdot(y-z)}{(w-z)^2\,(y-z)^2}\right.$$
$$\left.-\,\frac{(y-w)^i}{(y-w)^2}\,\frac{(w-z)\cdot(x-z)}{(w-z)^2\,(x-z)^2}\right]\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\}\Big]$$
$$-\,\frac{1}{(2\pi)^3}\,\frac{i\,g^4\,f^{hbc}}{32\pi^4}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x'}\frac{(x'-w')^i}{(x'-w')^2}\left[\frac{(x-w)^i}{(x-w)^2}\,\log\frac{(w-y)^2}{(x-y)^2}-\frac{(y-w)^i}{(y-w)^2}\,\log\frac{(w-x)^2}{(x-y)^2}\right]\times$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\}\Big] \tag{2.1}$$

This can be written as follows:

$$\begin{aligned}\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[&\\
-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')&\\
+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^b(x')\,\rho^e(x)&\\
+\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')&\\
-\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)&\\
+\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w')\,\rho^c(x)\,\rho^e(x')&\\
+\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)&\\
-\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ha}(w')\,\rho^c(x)\,\rho^e(x')\Bigg]&\end{aligned} \tag{2.2}$$

2.3 Row III

$$\begin{aligned}\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{&\\
-\,\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\big[(x-z)^2+(x-w)^2\big]}{\big[(x-z)^2-(x-w)^2\big]^2}&\\
\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\big]&\\
+\,\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\big[(x-w)^2+(x-z)^2\big]}{\big[(x-z)^2-(x-w)^2\big]^2}&\\
\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\big]&\\
+\,\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}&\\
\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\big]&\\
-\,\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}&\\
\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\big]\Bigg\}&\end{aligned} \tag{2.3}$$

$$
\begin{aligned}
\frac{d^3 N_{2\,\mathrm{NLO}}}{d^3 k} = &-\frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ \\
&-\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,(x-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\big] \\
&+\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\big] \\
&-\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\big] \\
&+\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\Big(\frac{\vee}{k^+}\Big)\,\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\big] \\
&-\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\Big(\frac{\vee}{k^+}\Big)\,\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(z-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\big] \\
&+\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\Big(\frac{\vee}{k^+}\Big)\,\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\big] \\
&-\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\Big(\frac{\vee}{k^+}\Big)\,\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-z)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\big] \\
&+\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\Big(\frac{\vee}{k^+}\Big)\frac{(x'-w')\cdot(x-z)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\big] \\
&-\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\Big(\frac{\vee}{k^+}\Big)\,\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\big]\Bigg\}
\end{aligned}
\tag{2.4}
$$

This can be written as follows:

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = {} & \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
& -\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
& +\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
& +\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
& -\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
& +\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
& +\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
& -\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
& -\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
& -\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
& +\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x) \\
& -N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
& +\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
& -\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x) \\
& -\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x') \\
& +\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x) \\
& -N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
& +N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
& +\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x) \\
& -\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(z)\,\rho^l(x)\,\rho^h(x') \\
& -\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x) \\
& +\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x') \\
& -\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
& +N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
& +\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
& -N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
& -N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
& -\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
& +\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x) \\
& +\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')\Bigg]
\end{aligned}
\tag{2.5}
$$

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[$$
$$\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)$$
$$+\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)$$
$$-\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$$
$$-\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^l(x')\,\rho^h(x)$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)\Bigg] \tag{2.6}$$

2.4 Row IV

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)}\Bigg\{$$
$$\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.$$
$$\left.-\,\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2}+\frac{\delta^{ij}}{2\,(z-w)^2}\right]$$
$$-\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}$$
$$+\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^e(y)\rho^{c'}(x)-f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^e(y)\rho^d(x)\Big] \tag{2.7}$$

This can be written as:

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x) \\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x') \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x) \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x) \\
+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x) \\
+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x') \\
-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x) \\
-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x) \\
+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Bigg]
\end{aligned}
\tag{2.8}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
+\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x) \\
-\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x) \\
+\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x) \\
+\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^m(x')\,\rho^e(x) \\
-\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x) \\
-\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^m(x')\,\rho^e(x)\Bigg]
\end{aligned}
\tag{2.9}
$$

2.5 Row V

First part

NA

Second part

NA

3 Group II

3.1 Row I

3.1.1 First part

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\Bigg\{ \\
-\;\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2} \\
\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right] \\
+\;\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2} \\
-\;\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\} \\
\left[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^b(y)\}-U^{c'b}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right]
\end{aligned}
\tag{3.1}
$$

$$
\begin{aligned}
\frac{d^3 N_{1\,\mathrm{NLO}}}{d^3 k} = \frac{1}{(2\pi)^3} \int_{x,y,z,w,x',w'} e^{-ik(w'-w)} \Bigg\{ & \\
& \frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\,\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2} \\
& \left[\frac{\delta^{ij}}{2\,(z-w')^2} + \frac{(x-z)^j\,(z-w')^i}{(z-w')^2\,(x-z)^2} - \frac{(x-z)^j\,(x-w')^i}{(x-w')^2\,(x-z)^2} - \frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}\right] \\
& - \frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\,\frac{(x'-w)\cdot(y-z)}{(x'-w)^2\,(y-z)^2\,(z-w')^2} \\
& + \frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\,\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\left[\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2} + \frac{(x-z)^j\,(x-w')^i}{2\,(x-z)^2\,(x-w')^2}\right] \Bigg\} \\
& \times \left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x) - f^{ac'd'}\,U^{c'b}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\right] \times \left[U^{ac}(w)\,\{\rho^c(x'),\rho^b(y)\} - U^{cb}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^d(x'),\rho^e(y)\}\right]
\end{aligned}
\tag{3.2}
$$

We can write it as follows:

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x) \\
-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x) \\
+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ha}(w')\,\rho^c(x)\,\rho^e(x') \\
+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x) \\
+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(z)\,\rho^l(x)\,\rho^h(x') \\
-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x) \\
-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x) \\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x) \\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^c(x)\,\rho^b(x') \\
+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^l(x')\,\rho^h(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^m(x')\,\rho^e(x) \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\Bigg]
\end{aligned}
\tag{3.3}
$$

3.1.2 Second part

$$
\begin{aligned}
\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times \\
&\Bigg[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2} \\
&+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\Bigg]\times\Bigg[(x-z)^2-\frac{(x-w)^2\,(x'-z)^2}{(x'-w')^2}\Bigg]\times \\
&\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x) \\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg] \\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times \\
&\Bigg[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2} \\
&+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\Bigg] \\
&\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x) \\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg] \\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-z)^2}{[(x'-w')^4\,(x-z)^2\,-(x-w)^2\,(x'-z)^2\,(x'-w')^2]}\times \\
&\Bigg[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2} \\
&+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\Bigg] \\
&\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x) \\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg]
\end{aligned}
\tag{3.4}
$$

$$
\begin{aligned}
\frac{d^3 N_{2c\,\mathrm{NLO}}}{d^3 k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2}{(x'-z)^2}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]} \\
&\Bigg[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2} \\
&+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\Bigg] \\
&\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x) \\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg] \\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-w)^2}{(x-z)^2}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]} \\
&\Bigg[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2} \\
&+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\Bigg] \\
&\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x) \\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg]
\end{aligned}
\tag{3.5}
$$

$$
\begin{aligned}
\frac{d^3N_{2d\,\mathrm{NLO}}}{d^3k} &= -\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2-(x'-z)^2]} \\
&\Bigg[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2} \\
&+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\Bigg] \\
&\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x) \\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg] \\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]} \\
&\Bigg[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2} \\
&+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\Bigg]\times \\
&\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x) \\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg] \\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-z)^2}{[(x'-w')^2-(x'-z)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]} \\
&\Bigg[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2} \\
&+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\Bigg] \\
&\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x) \\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg]
\end{aligned}
\tag{3.6}
$$

$$
\begin{aligned}
\frac{d^3N_{2e\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2-(x'-z)^2]}\\
&\Bigg[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\\
&+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\Bigg]\\
&\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-w)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\Bigg[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\\
&+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\Bigg]\times\\
&\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2}{[(x'-w')^2-(x'-z)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\Bigg[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\\
&+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\Bigg]\\
&\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg]
\end{aligned}
\tag{3.7}
$$

$$
\begin{aligned}
\frac{d^3N_{2f\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\log\Big(\frac{\vee}{k^+}\Big)\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{\big[(x'-w')^2\,(x-w)^2-(x-z)^2\,(x'-z)^2\big]}{(z-w)^2\,(z-w')^2\,\big[(x-z)^2-(x-w)^2\big]\,\big[(x'-z)^2-(x'-w')^2\big]} \\
&\Big[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x')-f^{ac'd'}\,U^{c'b}(z)U^{d'e'}(x')\,\rho^{e'}(x')\Big]\times\,\big[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^{e}(x)\big] \\
&-\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2\,(x-w)^2}{(z-w)^2\,(z-w')^2\,\big[(x-z)^2-(x-w)^2\big]} \\
&\frac{\big[(x'-z)^2-(x'-w')^2\big]}{\big[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\big]} \\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x) \\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big] \\
&+\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2\,(x'-z)^2}{(z-w)^2\,(z-w')^2} \\
&\frac{\big[(x-z)^2-(x-w)^2\big]}{\big[(x'-z)^2-(x'-w')^2\big]\,\big[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\big]} \\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x) \\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]
\end{aligned}
\tag{3.8}
$$

We can write it as:

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x') \\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x') \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x') \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x) \\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{c}(x)\,\rho^{e}(x') \\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{c}(x')\,\rho^{e}(x) \\
&-\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x')\,\rho^{c}(x)\,\rho^{l}(x') \\
&-\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x) \\
&-\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(x)\,\rho^{l}(x)\,\rho^{h}(x') \\
&-\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(x')\,\rho^{l}(x')\,\rho^{h}(x) \\
&+2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ba}(x)\,U^{bc}(x')\,\rho^{a}(x)\,\rho^{c}(x') \\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x) \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x') \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x) \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{c}(x')\,\rho^{e}(x)\Big]
\end{aligned}
\tag{3.9}
$$

3.2 Row II

NA

3.3 Row III

NA

3.4 Row IV

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}$$
$$\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\times\left[U^{ce}(z)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x) \tag{3.10}$$

This can be written as:

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[&\\ +2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ba}(x)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')&\\ +2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,\rho^a(x)\,\rho^a(x')&\\ -2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ba}(z)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')&\\ -2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ba}(z)\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\Bigg]&\end{aligned} \tag{3.11}$$

4 Group III (with c.c.)

4.1 Row I

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\Bigg\{&\\ -\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\Big(\frac{\vee}{k^+}\Big)\times\Bigg[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}&\\ -\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\Bigg]&\\ +\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}&\\ -\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\,\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\Bigg[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\Bigg]\Bigg\}\times&\\ \Big[U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,\rho^{b'}(x')\rho^{d'}(y)-U^{d'b}(z)\,U^{d'e'}(y)U^{ac'}(w')\,\rho^{c'}(x')\rho^{e'}(y)\Big]\times&\\ \Big[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\Big]&\end{aligned} \tag{4.1}$$

This can be written as:

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x) \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x) \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x') \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x) \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x) \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x) \\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x) \\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x') \\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x) \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')\Bigg]
\end{aligned}
\tag{4.2}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x) \\
-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
+\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
-\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
-\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x) \\
+\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x) \\
-\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x) \\
-\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^m(x')\,\rho^e(x) \\
+\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x) \\
+\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^m(x')\,\rho^e(x)\Bigg]
\end{aligned}
\tag{4.3}
$$

4.2 Row II

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\Bigg\{ \\
-\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\Big(\frac{\vee}{k^+}\Big)\times\Bigg[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2} \\
-\,\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2}\Bigg] \\
+\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2} \\
-\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\,\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\Bigg[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\Bigg]\Bigg\}\times \\
\Bigg[U^{c'd'}(y)\,U^{ab'}(x')\,U^{c'b}(z)\,\rho^{d'}(y)\rho^{b'}(x')-U^{ab'}(x')\,\rho^b(y)\rho^{b'}(x')\Bigg]\times\Bigg[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\Bigg]
\end{aligned}
\tag{4.4}
$$

This can be written as:

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x') \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^b(x')\,\rho^e(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x) \\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x) \\
+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x) \\
-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w')\,\rho^c(x)\,\rho^e(x') \\
+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
-2\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x) \\
+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')\Bigg] \qquad (4.5)
\end{aligned}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
&- 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x') \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x) \\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^b(x)\,\rho^c(x') \\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x) \\
&- 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x) \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^m(x')\,\rho^e(x) \\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(x)\,U^{ha}(w')\,\rho^c(x')\,\rho^e(x)\Bigg]
\end{aligned}
\tag{4.6}
$$

4.3 Row III

4.3.1 First part

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = &-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2} \\
&\Big[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\big] \\
&+\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2} \\
&\Big[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\big] \\
&-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,x',w',z,y,x} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(w'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2} \\
&\Big[U^{c'e'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\big]
\end{aligned}
\tag{4.7}
$$

We can write it as:

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
+ N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
+ N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x) \\
- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
- N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
+ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x) \\
+ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{he}(z)\,\rho^d(x)\,\rho^c(x') \\
+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(z)\,\rho^b(x)\,\rho^c(x') \\
+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x) \\
- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^l(x')\,\rho^h(x) \\
- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(w)\,U^{he}(z)\,\rho^d(x')\,\rho^c(x)\Bigg]
\end{aligned}
\tag{4.8}
$$

4.3.2 Second part

$$
\begin{aligned}
\frac{d^3N_{2\,\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\Bigg[\frac{(x-w)\cdot(x'-y')\,(z-w)\cdot(y'-z)}{(z-w)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2} \\
+ \frac{(x-z)\cdot(y'-z)\,(x-w)\cdot(x'-y')}{2\,(x-z)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\Bigg]\,\Big[f^{c'd'a}\,U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times \\
\big[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\big] \\
+ \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\Bigg[\frac{(x-w)\cdot(x'-y')\,(z-w)\cdot(y'-z)}{(z-w)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2} \\
+ \frac{(x-z)\cdot(y'-z)\,(x-w)\cdot(x'-y')}{2\,(x-z)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\Bigg]\,\Big[f^{c'd'a}\,U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times \\
\big[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\big] \\
- \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,x',w',z,x} e^{-ik\cdot(w'-w)}\,\frac{(x'-z)\cdot(x-z)}{(x'-z)^2\,(x-z)^2}\,\Bigg[\frac{(w'-z)\cdot(z-w)}{(w'-z)^2\,(z-w)^2} - \frac{(w'-z)\cdot(x-w)}{2\,(w'-z)^2\,(x-w)^2}\Bigg] \\
\Big[f^{c'd'a}\,U^{c'e'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\big]
\end{aligned}
\tag{4.9}
$$

This can be written as:

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
&+\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x')\,\rho^c(x)\,\rho^l(x') \\
&+\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x) \\
&+\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x') \\
&+\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x) \\
&-\,4\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ba}(x)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x') \\
&+\,2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ba}(z)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x') \\
&+\,2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ba}(z)\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x) \\
&+\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w)\,U^{he}(z)\,\rho^d(x)\,\rho^c(x') \\
&+\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w)\,U^{he}(z)\,\rho^d(x')\,\rho^c(x) \\
&+\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{he}(z)\,\rho^c(x)\,\rho^d(x') \\
&+\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{he}(z)\,\rho^c(x')\,\rho^d(x) \\
&+\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
&+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{he}(z)\,\rho^d(x)\,\rho^c(x') \\
&+\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&-\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x) \\
&-\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(w)\,U^{he}(z)\,\rho^d(x')\,\rho^c(x)\Bigg]
\end{aligned}
\tag{4.10}
$$

4.4 Row IV

NA

4.5 Row V

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = &-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2} \\
&\Big[U^{c'e'}(y')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\big] \\
&+\,\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2} \\
&\Big[U^{c'e'}(y')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\big] \\
&-\,\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,w',z,y,x,x'} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(w'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2} \\
&\Big[U^{c'e'}(z)\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\,\big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\big]
\end{aligned}
\tag{4.11}
$$

This can be written as:

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
-\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
+\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
+\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x) \\
-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
-\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
-\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
+\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
+\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
+\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
-\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x) \\
+\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^l(x')\,\rho^h(x) \\
+\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)\Bigg]
\end{aligned}
\tag{4.12}
$$

4.6 Row VI

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = -\,\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2} \\
\Big[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\big] \\
+\,\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2} \\
\Big[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\big] \\
-\,\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,w',z,y,x,x'} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(w'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2} \\
\Big[U^{c'e'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\,\big[U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\big]
\end{aligned}
\tag{4.13}
$$

This can be written as:

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
+ 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
- N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
- N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x) \\
- 2\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
+ 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
+ 2\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
- 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{he}(z)\,\rho^d(x)\,\rho^c(x') \\
- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(z)\,\rho^b(x)\,\rho^c(x') \\
- 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x) \\
+ 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(x)\,U^{he}(z)\,\rho^d(x')\,\rho^c(x)\Bigg]
\end{aligned}
\tag{4.14}
$$

5 BFKL contribution

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{k^+}\Big]\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[
&N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^b(x')\,\rho^e(x)\\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(x)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x)\\
&+ \mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x)\,\rho^e(x')\\
&+ \mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&+ 2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,\rho^a(x)\,\rho^a(x')\\
&+ \mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w)\,U^{he}(z)\,\rho^d(x)\,\rho^c(x')\\
&+ \mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w)\,U^{he}(z)\,\rho^d(x')\,\rho^c(x)\\
&+ \mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{he}(z)\,\rho^c(x)\,\rho^d(x')\\
&+ \mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{he}(z)\,\rho^c(x')\,\rho^d(x)\\
&- N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+ N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+ N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&- N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&- 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)\\
&- 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+ 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{he}(z)\,\rho^d(x)\,\rho^c(x')\\
&- 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{he}(z)\,\rho^d(x)\,\rho^c(x')\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^c(x)\,\rho^b(x')\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^b(x)\,\rho^c(x')\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(z)\,\rho^b(x)\,\rho^c(x')\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(z)\,\rho^b(x)\,\rho^c(x')\\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(x)\,U^{ha}(w')\,\rho^c(x')\,\rho^e(x)\\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(w)\,U^{he}(z)\,\rho^d(x')\,\rho^c(x)\\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(x)\,U^{he}(z)\,\rho^d(x')\,\rho^c(x)\Bigg]
\end{aligned}
\tag{5.1}
$$

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{k^+}\Big]\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\Big[\\
- N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) + N_c\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
+ N_c\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) - N_c\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\Big] \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\Big[\\
- N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) + N_c\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
+ N_c\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) - N_c\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\Big] \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big[\\
+ 2\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x) - 2\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x) \\
- 2\,f^{abc}\,f^{dbe}\,U^{ha}(x')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x) + 2\,f^{abc}\,f^{dbe}\,U^{ha}(x')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)\Big] \\
+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big[\\
+ 2\,N_c\,\rho^a(x)\,\rho^a(x') - \,f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(w)\,\rho^b(x)\,\rho^e(x') \\
- \,f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x) + \,f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(z)\,\rho^b(x)\,\rho^e(x') + \,f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(z)\,\rho^b(x')\,\rho^e(x) \\
+ \,f^{abc}\,f^{dce}\,U^{ha}(z)\,U^{hd}(w)\,\rho^b(x)\,\rho^e(x') + \,f^{abc}\,f^{dce}\,U^{ha}(z)\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x)\Big] \\
+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\Big[\\
- N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) + N_c\,U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x) + N_c\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
- N_c\,U^{ab}(x')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\Big] \\
+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big[\\
+ 2\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x') - 2\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(z)\,\rho^b(x)\,\rho^e(x') \\
- 2\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x') + 2\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(z)\,\rho^b(x)\,\rho^e(x')\Big] \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\Big[\\
- N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^c(x)\,\rho^b(x') + N_c\,U^{ab}(w')\,U^{ac}(x)\,\rho^c(x)\,\rho^b(x') \\
+ N_c\,U^{ab}(z)\,U^{ac}(w)\,\rho^c(x)\,\rho^b(x') - N_c\,U^{ab}(z)\,U^{ac}(x)\,\rho^c(x)\,\rho^b(x')\Big] \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big[\\
+ 2\,f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x) - 2\,f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{hd}(x)\,\rho^b(x')\,\rho^e(x) \\
- 2\,f^{abc}\,f^{cde}\,U^{ha}(z)\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x) + 2\,f^{abc}\,f^{cde}\,U^{ha}(z)\,U^{hd}(x)\,\rho^b(x')\,\rho^e(x)\Big]\Bigg] \qquad (5.2)
\end{aligned}
$$

In color singlet form, we get

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{1}{N_c^2-1}\,\log\Big[\frac{\vee}{k^+}\Big]\int_{x,y,z,w,w'} e^{-ik(w'-w)}\,\langle\rho^f(x)\rho^f(y)\rangle\,\Bigg[$$
$$\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[2N_c{\rm Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big]$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[-N_c{\rm Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big]$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[-N_c{\rm Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big]$$
$$+\,K(w,w',z)\,K(x,y,z)\Big[2N_c{\rm Tr}\{(U^\dagger(z)-U^\dagger(w))(U(z)-U(w'))\}\Big]$$
$$+\,\mathcal{K}(x-w')\cdot\mathcal{K}(w-z)\,K(x,y,z)\Big[2N_c{\rm Tr}\{(U^\dagger(z)-U^\dagger(w))(U(x)-U(w'))\}\Big]$$
$$+\,\mathcal{K}(x-w')\cdot\mathcal{K}(w-z)\,K(y,y,z)\Big[-N_c{\rm Tr}\{(U^\dagger(z)-U^\dagger(w))(U(x)-U(w'))\}\Big]$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,K(x,y,z)\Big[2N_c{\rm Tr}\{(U^\dagger(x)-U^\dagger(w))(U(z)-U(w'))\}\Big]$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,K(y,y,z)\Big[-N_c{\rm Tr}\{(U^\dagger(x)-U^\dagger(w))(U(z)-U(w'))\}\Big]\Bigg] \tag{5.3}$$

where $K(x,y,z)=\frac{(x-z)\cdot(y-z)}{(x-z)^2\,(y-z)^2}$